Shreya H S

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Professional Summary

Artificial Intelligence and Machine Learning graduate (B.E., CGPA 8.4/10) with hands-on experience building Generative AI and machine learning applications, including LLM-based retrieval-augmented generation (RAG) systems and an agentic AI voice agent. Proficient in Python and Java, with experience across the development cycle from data preprocessing and API integration to testing and deployment. Seeking a Junior AI Development role building LLM-powered products.

Education

Acharya Institute of Technology 2022 – 2026
Bachelor of Engineering in Artificial Intelligence and Machine Learning *Bengaluru, Karnataka*
- CGPA: 8.4 / 10

Experience

CloudFirst Jun 2026 – Aug 2026
Artificial Intelligence Research Analyst *Bengaluru, India*
- Built an agentic AI voice calling agent in Python as a layered system, separating a Sarvam AI voice layer (speech recognition and synthesis) from a business layer for knowledge retrieval and response logic.
- Implemented confidence-based escalation that routes low-confidence calls to a human relationship manager, achieving 90% routing accuracy on a 100-call evaluation set.
- Wrote automated tests and designed the business layer for AWS integration, making the agent integration-ready.

Tap Academy Jan 2026 – May 2026
Full Stack Developer Intern *Bengaluru, India*
- Built full-stack applications with Java, React, and MySQL, integrating front-end, back-end, and database layers.
- Collaborated with senior developers through code reviews, writing maintainable code.

IBM SkillsBuild / Edunet Foundation
Artificial Intelligence and Cloud Intern *Remote*
- Built and deployed a Large Language Model (LLM) chatbot on IBM Cloud using watsonx.ai.
- Implemented data preprocessing, feature engineering, and model optimisation in IBM Watson Studio.

Projects

AI Voice Calling Agent | *Python, FastAPI, RAG, Sarvam AI, React, TypeScript* GitHub
- Built a FastAPI backend with provider adapters (Sarvam AI speech, Twilio, Amazon Bedrock), grounded RAG over an 18-document knowledge base, and pre-LLM compliance and prompt-injection guardrails.
- Implemented five-factor confidence scoring and five-tier escalation to a human Relationship Manager, with CRM REST APIs (customer profiles, lead qualification, follow-ups) and a React dashboard.
- Validated with 154 automated pytest unit, integration, and security tests; documented as a proof of concept.

HealthDoc Assistant | *Python, LangChain, FAISS, Ollama (Mistral), Streamlit* GitHub
- Developed a private, offline RAG application for question answering over medical documents, achieving 93% grounded-answer rate and 95% top-5 retrieval hit rate on a 100-question test set.
- Built the pipeline with document chunking, Ollama embeddings, FAISS vector search, and a local Mistral LLM behind a multi-turn Streamlit interface, keeping documents on the user's machine.

Emotionally Intelligent Blockchain Security | *Python, TensorFlow, OpenCV, Blockchain*
- Engineered a real-time emotion-recognition CNN (92% accuracy) integrated with blockchain-backed IoT security.
- Implemented tamper-resistant audit logging and adaptive threat response; evaluated machine learning intrusion detection on NSL-KDD, achieving 95% Random Forest accuracy.

Technical Skills

Programming: Python, Java, SQL
AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Agentic AI, CNNs
LLM Development: RAG, Prompt Engineering, Embeddings, Vector Search, LangChain, FAISS, Ollama
Frameworks and Tools: Scikit-learn, TensorFlow, OpenCV, FastAPI, Streamlit, Pytest
Development: REST APIs, React, HTML, CSS, JDBC, Servlets, Git, GitHub, VS Code
Databases: MySQL, MongoDB
Cloud: IBM Cloud, IBM watsonx.ai, IBM Watson Studio, AWS integration

Certifications

Machine Learning Specialization and **Generative AI for Everyone** – Coursera (DeepLearning.AI)
Python for Everybody – Coursera (University of Michigan) | **Data Science for Engineers** – NPTEL